EV Charging Behavior Analysis and Financial Risk

Dataset Details

Dataset Name: EV Charging Behavior Analysis and Financial Risk

Source Link: https://www.kaggle.com/datasets/nithinreddy600/ev-charging-behavior-analysis-and-financial-risk

Number of Rows and Columns: 18,941 rows and 18 columns

Missing Values: 0 missing values detected in the uploaded CSV

Target Class	Count	Percentage
Low/No High Default Risk (0)	10,530	55.59%
High Default Risk (1)	8,411	44.41%

Target Variable

The main target variable is High_Default_Risk, a binary variable indicating whether an EV charging user is at high financial default risk (1) or not at high default risk (0).

Charging_Efficiency_Index will be analyzed as an important supporting variable for understanding charging performance, charger working status, charging location, charging time, charging cost, and default-risk patterns.

Type of Task

Supervised Learning - Classification, focused on predicting High_Default_Risk.

Additional exploratory analysis will examine how charger working status and charging location relate to charging efficiency, charging time, charging cost, and risk rate.

Problem Statement

EV charging platforms increasingly collect financial, demographic, app-usage, vehicle, and charging-behavior data. This project aims to use those variables to predict whether an EV charging user is likely to become a high default risk. Accurately identifying high-risk users can support better financial decision-making, early risk detection, and more efficient charging-service management. The project will also examine whether charging behavior and product-performance variables, such as charger working status, charging time, charging cost, charging location, and charging efficiency, improve default-risk prediction beyond traditional demographic and financial variables alone.

Research Questions

1. How accurately can high default risk be predicted using EV-user financial, demographic, app-usage, and charging-behavior variables?
2. Which financial indicators are the strongest predictors of high default risk among EV charging users?
3. Does adding charging-behavior and product-performance features improve default-risk prediction beyond demographic and financial variables alone?
4. Which dataset features contribute most to machine-learning predictions of high default risk?
5. How do charger working status and charging location relate to charging efficiency, charging time, charging cost, and risk rate?

Proposed Methodology

1. Exploratory Data Analysis (EDA): Analyze the distribution of High_Default_Risk and examine numerical and categorical feature patterns, including missed payments, income level, loan status, app usage, charging cost, charging time, charging location, charger working status, and charging efficiency.
2. Data Quality Checks: Check missing values, duplicates, invalid ranges, outliers, and class balance. User_ID will be treated as an identifier and removed from model training.
3. Data Preprocessing: Encode categorical variables such as City_Tier, EV_Type, Income_Level, Charging_Location_Type, and Charger_Working_Status using one-hot encoding or ordinal encoding where appropriate.
4. Feature Engineering: Create grouped feature sets to compare financial/demographic variables against expanded models that include charging behavior and product-performance variables.
5. Train-Test Split: Split the dataset into training and testing sets, such as 80% training and 20% testing, using stratification for the classification target.
6. Scaling: Apply standardization to algorithms that are sensitive to scale, such as Logistic Regression, Support Vector Machine, and K-Nearest Neighbors.
7. Classification Modeling: Train and compare multiple supervised classification models to predict High_Default_Risk.
8. Feature Importance and Model Interpretation: Use Random Forest feature importance, permutation importance, and SHAP-style interpretation where available to identify the strongest predictors of high default risk.

Machine Learning Models to be Used

1. Logistic Regression - baseline classification model for predicting High_Default_Risk.
2. Decision Tree Classifier - interpretable classification model for identifying rule-based risk patterns.
3. Random Forest Classifier - ensemble model for non-linear classification and feature-importance analysis.
4. Gradient Boosting or XGBoost Classifier - advanced ensemble model for improving predictive performance.
5. Support Vector Machine or K-Nearest Neighbors - optional comparison models after scaling numerical features.

Evaluation Metrics

The classification models will be evaluated using the following metrics:

- Classification: Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Recall will be especially important because missing high-risk users may create greater financial risk.
- Model Comparison: A final table will compare model performance across baseline, financial-only, and full-feature classification models to test whether charging behavior improves default-risk prediction.

Expected Figures and Tables

The following figures and tables will be included in the final project report and directly match Research Questions 1-5.

RQ1: Classification performance for predicting high default risk

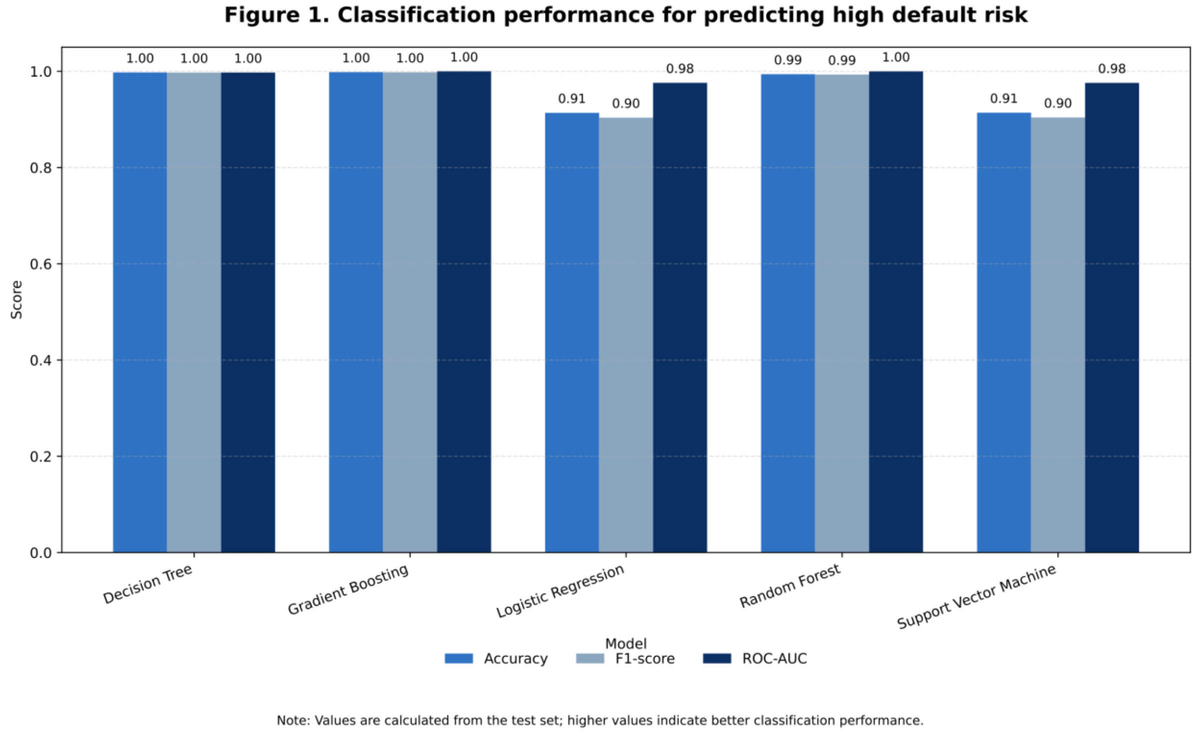

Figure 1. Classification performance for predicting high default risk

Note: Values are calculated from the test set; higher values indicate better classification performance.

Table 1. Classification performance results for high default-risk prediction

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gradient Boosting	0.9982	1	0.9958	0.9979	1
Random Forest	0.9939	0.9946	0.9917	0.9932	0.9999
Decision Tree	0.9976	0.9988	0.9958	0.9973	0.9974
Logistic Regression	0.9137	0.896	0.9115	0.9037	0.976
Support Vector Machine	0.914	0.8956	0.9127	0.9041	0.9759

Note: The table compares classification models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

RQ2: Strongest financial predictors of high default risk

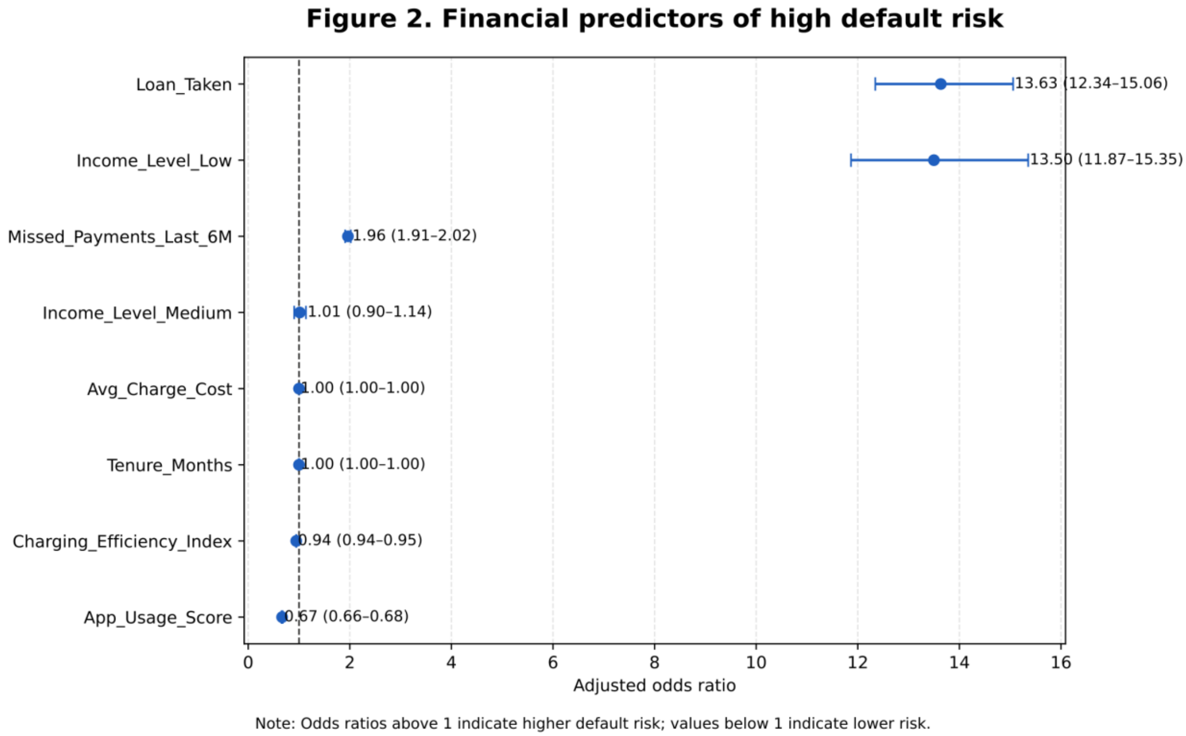

Figure 2. Financial predictors of high default risk

Note: Odds ratios above 1 indicate higher default risk; values below 1 indicate lower risk.

Table 2. Financial predictor odds ratios for high default risk

Predictor	Log Odds	Adjusted Odds Ratio	CI Lower	CI Upper	P-Value	Direction
Loan_Taken	2.6126	13.634	12.3449	15.0577	0	Increases risk
Income_Level_Low	2.6026	13.4984	11.8667	15.3545	0	Increases risk
Missed_Payments_Last_6M	0.6746	1.9632	1.908	2.0199	0	Increases risk
Income_Level_Medium	0.0142	1.0143	0.905	1.1368	0.8073	Increases risk
Avg_Charge_Cost	0.0001	1.0001	0.9997	1.0004	0.6001	Increases risk
Tenure_Months	-0.0002	0.9998	0.9967	1.0028	0.8786	Reduces risk
Charging_Efficiency_Index	-0.0566	0.945	0.9428	0.9472	0	Reduces risk
App_Usage_Score	-0.4053	0.6668	0.6555	0.6782	0	Reduces risk

Note: Odds ratios and confidence intervals summarize the relationship between financial indicators and high default risk.

RQ3: Added value of charging-behavior and product-performance features

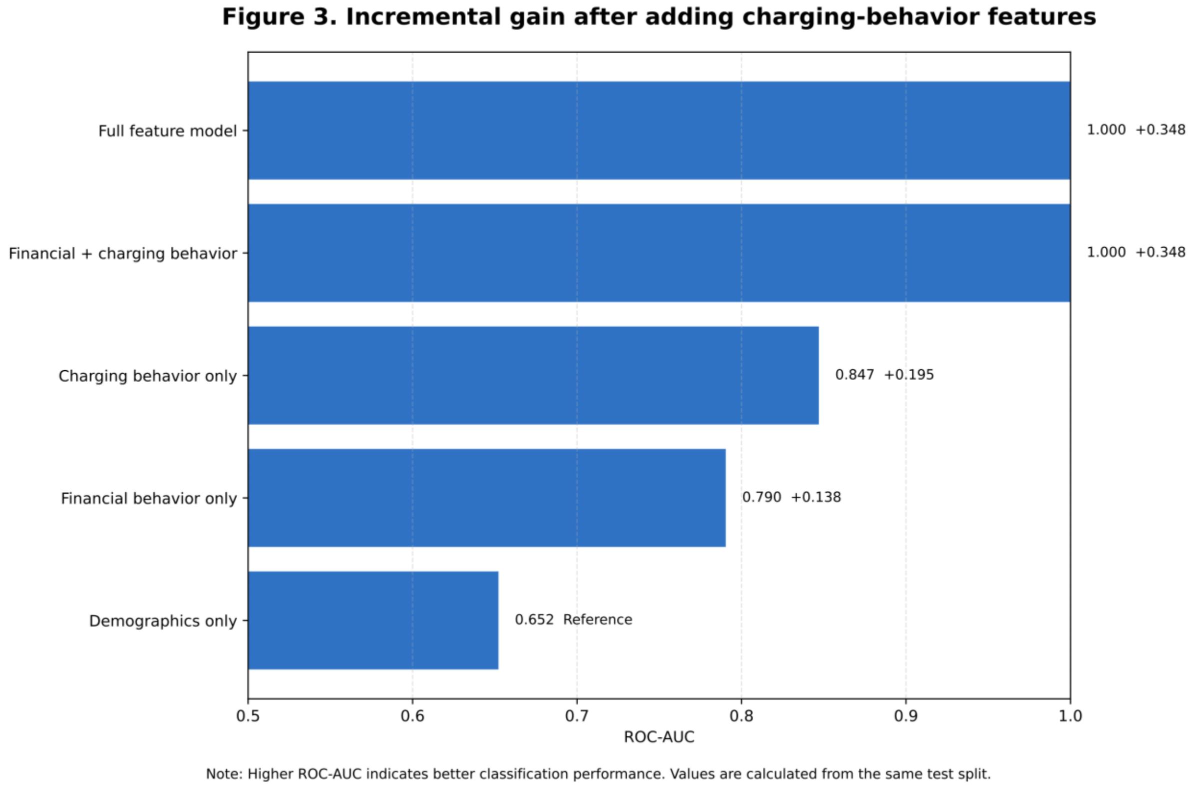

Figure 3. Incremental gain after adding charging-behavior features

Note: Higher ROC-AUC indicates better classification performance. Values are calculated from the same test split.

Table 3. Feature block ablation results

Feature Block Model	Variables Included	Accuracy	F1-Score	ROC-AUC	Delta ROC-AUC vs. Demographics
Demographics only	Age, City_Tier, Income_Level	0.6353	0.5753	0.6522	0
Financial behavior only	Income_Level, Loan_Taken, Missed_Payments_Last_6M, Tenure_Months, Avg_Charge_Cost	0.7271	0.6884	0.7904	0.1381
Charging behavior only	EV_Type, Battery_Capacity_kWh, Charging_Sessions_Per_Month, Distance_Travelled_Per_Month, Charging_Location_Type, Charger_Working_Status, Charging_Time_Minutes, Charging_Efficiency_Index, App_Usage_Score	0.7627	0.7177	0.847	0.1947
Financial + charging behavior	Income_Level, Loan_Taken, Missed_Payments_Last_6M, Tenure_Months, Avg_Charge_Cost, EV_Type, Battery_Capacity_kWh, Charging_Sessions_Per_Month, Distance_Travelled_Per_Month, Charging_Location_Type, Charger_Working_Status, Charging_Time_Minutes, Charging_Efficiency_Index, App_Usage_Score	0.9931	0.9923	0.9999	0.3476
Full feature model	Age, City_Tier, EV_Type, Battery_Capacity_kWh, Charging_Sessions_Per_Month, Avg_Charge_Cost, Distance_Travelled_Per_Month, Income_Level, Loan_Taken, Missed_Payments_Last_6M, Tenure_Months, Charging_Location_Type, App_Usage_Score, Charger_Working_Status, Charging_Time_Minutes, Charging_Efficiency_Index	0.9939	0.9932	0.9999	0.3477

Note: The ablation table compares model performance after adding different feature blocks.

RQ4: Most important machine-learning predictors

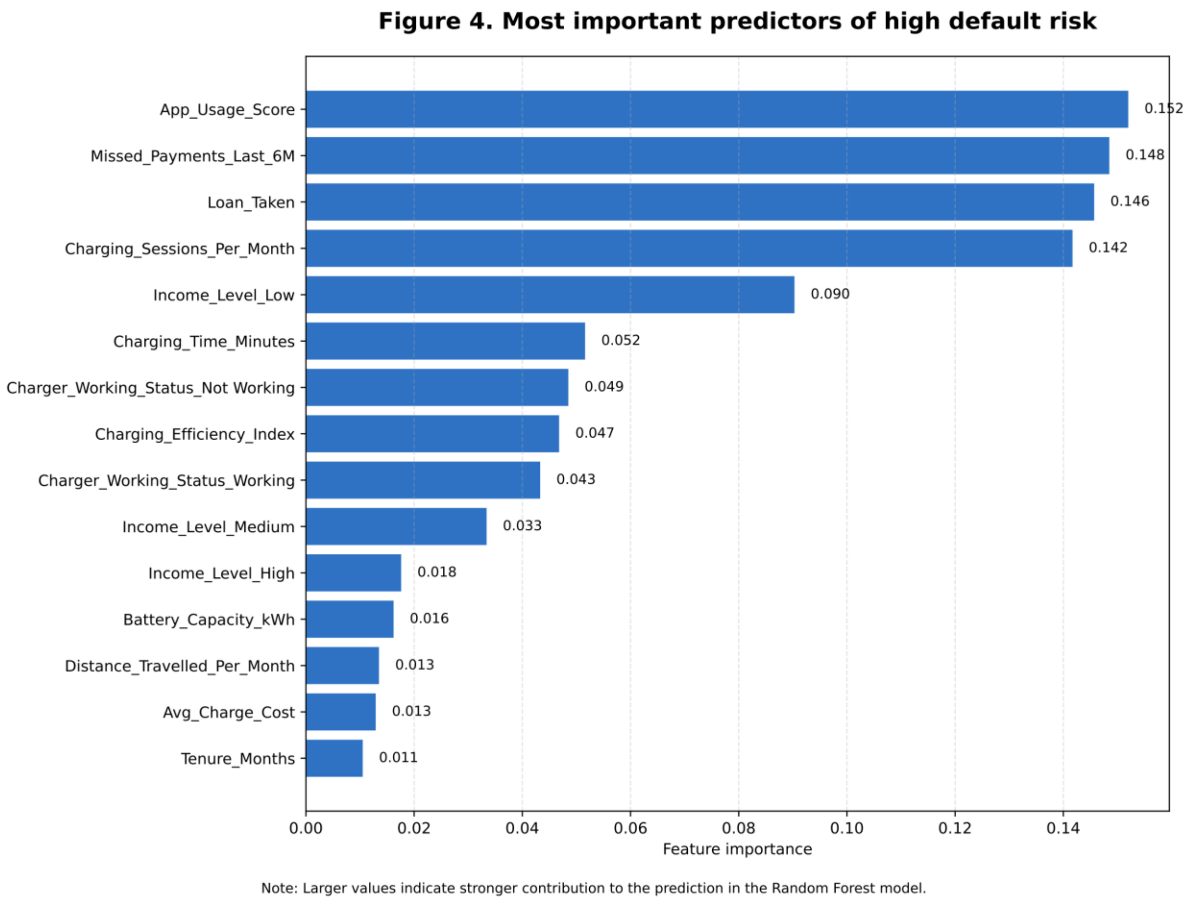

Figure 4. Most important predictors of high default risk

Note: Larger values indicate stronger contribution to the prediction in the Random Forest model.

Table 4. Feature importance ranking from the Random Forest model

Rank	Feature	Importance	Direction of Association
1	App_Usage_Score	0.152	Higher/present → lower risk
2	Missed_Payments_Last_6M	0.1485	Higher/present → higher risk
3	Loan_Taken	0.1457	Higher/present → higher risk
4	Charging_Sessions_Per_Month	0.1417	Higher/present → higher risk
5	Income_Level_Low	0.0903	Higher/present → higher risk
6	Charging_Time_Minutes	0.0516	Higher/present → higher risk
7	Charger_Working_Status_Not Working	0.0485	Higher/present → higher risk
8	Charging_Efficiency_Index	0.0468	Higher/present → lower risk
9	Charger_Working_Status_Working	0.0433	Higher/present → lower risk
10	Income_Level_Medium	0.0334	Higher/present → lower risk
11	Income_Level_High	0.0176	Higher/present → lower risk
12	Battery_Capacity_kWh	0.0162	Higher/present → higher risk
13	Distance_Travelled_Per_Month	0.0135	Higher/present → lower risk
14	Avg_Charge_Cost	0.0129	Higher/present → lower risk
15	Tenure_Months	0.0105	Higher/present → lower risk

Note: Feature importance values summarize each variable's contribution to the Random Forest prediction model.

RQ5: Charger working status, charging location, efficiency, cost, time, and risk rate

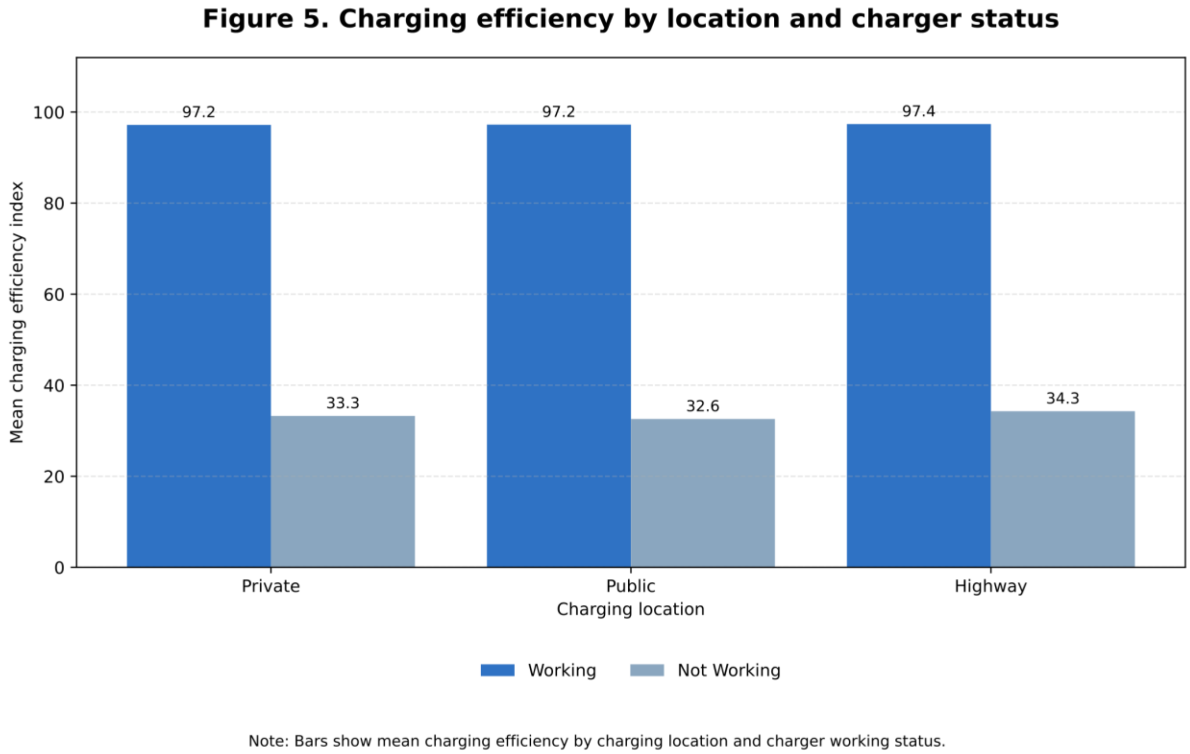

Figure 5. Charging efficiency by location and charger status

Note: Bars show mean charging efficiency by charging location and charger working status.

Table 5. Charging location and charger status summary

Charging Location Type	Charger Working Status	N Users	Mean Efficiency Index	Mean Charging Time (Minutes)	Mean Charge Cost	High Default Risk (%)
Highway	Not Working	1050	34.29	119.48	446.48	88.67
Highway	Working	5397	97.36	52.48	451.02	38.11
Private	Not Working	886	33.26	121.06	445.8	87.58
Private	Working	5307	97.18	52.7	451.4	34.82
Public	Not Working	914	32.59	123.44	446.87	88.73
Public	Working	5387	97.24	52.67	450.16	36.9

Note: The table summarizes charging performance and risk rate by charging location and charger working status.

Github Link

https://github.com/NithinReddy006/EV-charging-behavior-analysis-and-financial-risk.git